

Infosys Springboard Virtual Internship 6.0 Completion Report

Team Details <Do not mention any personally identifiable information like email ID, institute details, mobile phone number etc.>

Batch Number : 6

Start date : 10-11-2025

Name : Abishek.S

Internship Duration : 8 Weeks

1. Project Title

NeuroFleetX: Intelligent Taxi Dispatch System

2. Project Objective

The objective of the NeuroFleetX project was to design and develop a scalable and secure backend system for an intelligent taxi dispatch and fleet management platform. The system aims to efficiently manage users, drivers, and ride requests while integrating AI and machine learning capabilities to enhance transparency, predict estimated arrival times (ETA), and improve overall ride handling. A key goal was to ensure seamless frontend–backend communication and implement robust security mechanisms..

3. Project description in detail

NeuroFleetX is an intelligent taxi dispatch and fleet management system focused on backend development and AI integration. My role in the project was primarily as a **Backend Developer**, where I was responsible for integrating both **backend services** and a **complete machine learning ETA prediction pipeline**.

The backend was developed using **Spring Boot** with a layered architecture consisting of controller, service, repository, and model layers. This design ensured

modularity, scalability, and maintainability. Secure RESTful APIs were developed to handle authentication, user management, driver operations, ride history access, and administrative analytics.

A project was Involved the implementation of **JWT-based authentication and role-based access control (RBAC)**, defining roles such as ADMIN, DRIVER, and USER to ensure secure and authorized access to system resources. Passwords were securely stored using BCrypt hashing, and proper CORS configurations were applied to prevent unauthorized access.

The project also involved **database design and management using PostgreSQL**, where relational schemas were created using JPA and Hibernate. Core entities such as User, Driver, Ride Request, Taxi (structure only), and ML-related data were modeled to maintain data integrity and consistency.

A major contribution of this project was the **design and implementation of a complete ML-based ETA prediction system** using the **Random Forest algorithm**.

- Reading historical trip data from CSV files
- Train-test split using an **80/20 ratio** with a fixed random seed
- Training a **Random Forest model with 100 decision trees**
- Evaluating model performance using **R², RMSE, and MAE**
- Persisting the trained model to disk
- Storing evaluation metrics in **PostgreSQL** for performance tracking
- Exposing prediction functionality via secure backend APIs

This ML system significantly improved ETA prediction accuracy compared to traditional linear approaches and was designed to be **robust, explainable, and production-ready**.

AI-Based Ride History Summarization:

- The project also integrated the **OpenAI API** to generate **human-readable summaries of ride history**, improving transparency and user experience by converting raw ride data into meaningful explanations.

Another key feature was the integration of the **OpenAI API** to generate AI-powered, human-readable summaries of ride history. This improved transparency and user experience by converting complex ride data into meaningful explanations. Seamless frontend interoperability was ensured by designing clear API contracts and verifying correct role-based behavior in the UI.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Requirement analysis and understanding the ETA prediction problem	Analyzed the ETA prediction problem and studied historical trip data requirements.
Week 2	ML pipeline design and baseline model selection	Designed the ML pipeline and implemented baseline ETA prediction using Linear Regression.
Week 3	Data preprocessing and feature engineering planning	Performed data preprocessing and feature engineering, including distance calculation and encoding.
Week 4	Model training and evaluation strategy definition	Trained and evaluated the Linear Regression model using an 80/20 train-test split.
Week 5	Model improvement and advanced algorithm selection	Identified model limitations and upgraded the system to a Random Forest-based ETA prediction model.
Week 6	Model optimization and performance evaluation	Trained and optimized the Random Forest model and evaluated it using R^2 , RMSE, and MAE.
Week 7	Backend integration planning for real-time predictions	Integrated the finalized ML model with the backend through a real-time prediction API.
Week 8	System testing, validation, and documentation	Conducted extensive testing, validation, and documentation of the ML system.

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Defined ETA prediction objectives, project scope, dataset requirements, and ML responsibilities.	12 Nov 2025
Baseline Model Development	Designed the ML pipeline and developed the initial ETA prediction model using Linear Regression.	24 Nov 2025
Mid-Term Review	Reviewed Linear Regression performance, identified limitations, refined features, and planned upgrade to Random Forest.	02 Dec 2025
Final Model Implementation	Implemented, optimized, and evaluated the Random Forest-based ETA prediction model using R^2 , RMSE, and MAE.	22 Dec 2025
Final Integration & Submission	Integrated the ML model with backend APIs, completed testing, documentation, and final submission.	02 Jan 2026

5b. Project execution details

1. Clone the Repository

The project source code was cloned from the GitHub repository using Git.

2. `git clone <repository-url>`

3. Open the Project

The backend project was opened in an IDE such as **IntelliJ IDEA** or **VS Code** with Java support.

4. Configure Application Properties

Database and security configurations were set in the `application.properties` /file, including

- PostgreSQL database URL, username, and password
- JWT secret and expiration settings
- OpenAI API key for AI-based ride history summarization

5. Database Setup

A PostgreSQL database was created, and tables were automatically generated using **JPA & Hibernate** on application startup.

6. Build the Project

Maven was used to resolve dependencies and build the application.

7. `mvn clean install`

8. Run the Backend Server

The Spring Boot application was started using:

9. `mvn spring-boot:run`

or by running the main application class.

10. API Testing and Verification

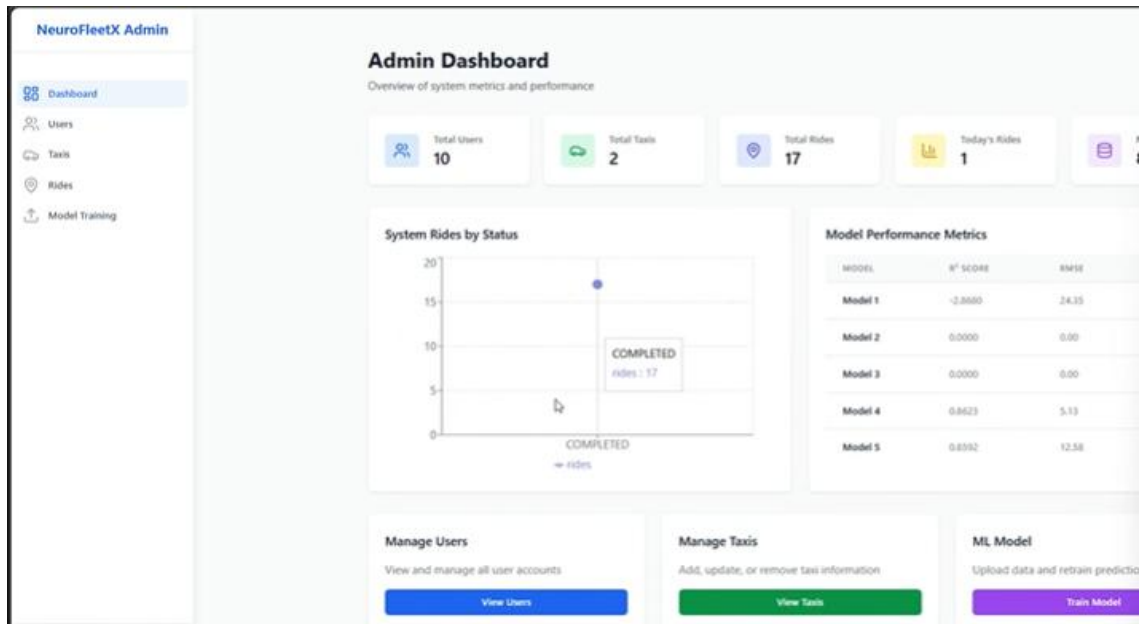
Backend APIs were tested using **Postman** to verify:

- Authentication and authorization
- Role-based access control (ADMIN, DRIVER, USER)
- Secure API responses and data integrity

11. Frontend Integration

The backend was connected to the frontend using REST APIs, ensuring smooth data exchange and role-based UI behavior.

6.Snapshots / Screenshots



```
NeuroFleetX > main > ETAPredictionService.java > ETAPredictionService.java
public class ETAPredictionService {
    public double predictETA(double originLat, double originLng, double destLat, double destLng, int usages, double trafficIndex, String weather, String timeOfDay) {
        if (!modelLoaded) {
            // If no model is loaded, try to load it, but return a default value if it fails
            try {
                loadModel();
            } catch (Exception e) {
                System.out.println("No trained model found, using default ETA: " + e.getMessage());
                // Return a default ETA based on distance
                double distance = calculateDistance(originLat, originLng, destLat, destLng);
                // Assuming average speed of 30 km/h, convert distance in km to minutes
                return Math.max(5.0, distance / 30.0 * 60.0); // Minimum 5 minutes
            }
        }
        double distance = calculateDistance(originLat, originLng, destLat, destLng);
        int weatherCode = weatherEncoding.getOrDefault(weather.toLowerCase(), defaultWeatherCode);
        int timeOfDayCode = timeOfDayEncoding.getOrDefault(timeOfDay.toLowerCase(), defaultTimeOfDayCode);
        System.out.println(distance);
        double[][] singleRow = {{distance, trafficIndex, (double)weatherCode, (double)timeOfDayCode, 80.0, 50.0}};
        String[] schema = {"distance", "trafficIndex", "weatherEncoded", "timeOfDayEncoded", "batteryLevel", "speed"};
        DataFrame inputData = DataFrame.of(singleRow, schema);
        // Formula formula = Formula.lhs("y");
        double[] predictions = ((smile.regression.RandomForest model).predict(inputData));
        return predictions[0];
    }
    private void saveModel() throws IOException {
        usages
    }
}
```

```

NeuroFleetXApplication.java
@ContextRoot("user_roles")
@EnableJpaRepositories(basePackages = "com.neurofleetx.repository")
public class NeuroFleetXApplication {

}

Run NeuroFleetXApplication
C:\Program Files\Java\jdk-17\bin\java.exe
...
:: Spring Boot :: (v3.1.0)

2026-01-12T15:23:57.544+05:30 INFO 15356 --- [main] com.neurofleetx.NeuroFleetXApplication : Starting NeuroFleetXApplication using Java 17.0.17 with PID 15356 (E
2026-01-12T15:23:57.553+05:30 INFO 15356 --- [main] com.neurofleetx.NeuroFleetXApplication : No active profile set, falling back to 1 default profile: "default"
2026-01-12T15:23:59.743+05:30 INFO 15356 --- [main] .s.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spring Data JPA repositories in DEFAULT mode.
2026-01-12T15:24:00.118+05:30 INFO 15356 --- [main] .s.d.r.c.RepositoryConfigurationDelegate : Finished Spring Data repository scanning in 301 ms. Found 6 JPA repos
2026-01-12T15:24:04.393+05:30 INFO 15356 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port(s): 8080 (http)
2026-01-12T15:24:04.441+05:30 INFO 15356 --- [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-01-12T15:24:04.442+05:30 INFO 15356 --- [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.8]
2026-01-12T15:24:04.828+05:30 INFO 15356 --- [main] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-01-12T15:24:05.834+05:30 INFO 15356 --- [main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 7086 ms
2026-01-12T15:24:05.269+05:30 INFO 15356 --- [main] o.hibernate.jpa.internal.util.LogHelper : HH0000204: Processing PersistenceUnitInfo [name: default]
2026-01-12T15:24:05.529+05:30 INFO 15356 --- [main] org.hibernate.Version : HH0000412: Hibernate ORM core version 6.2.2.Final
2026-01-12T15:24:05.537+05:30 INFO 15356 --- [main] org.hibernate.cfg.Environment : HH0000406: Using bytecode reflection optimizer
2026-01-12T15:24:05.955+05:30 INFO 15356 --- [main] o.h.b.i.BytecodeProviderInitiator : HH000021: Bytecode provider name : bytebuddy
2026-01-12T15:24:06.377+05:30 INFO 15356 --- [main] o.s.o.j.p.SpringPersistenceUnitInfo : No LoadTimeWeaver setup: ignoring JPA class transformer
2026-01-12T15:24:06.422+05:30 INFO 15356 --- [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...

```

```

pgAdmin 4
Object Explorer
neurofleetx1/postgres@PostgreSQL 17
model_metrics_id_seq
ride_request_id_seq
taxi_id_seq
users_id_seq
Tables (7)
  declined_ride
  driver_profiles
  model_metrics
    Columns (8)
      id
      mae
      r2
      rmse
      trained_at
      train_mae
      train_rmse
      train2
    Constraints (1)
      model_metrics_pkey
    Indexes
    RLS Policies
    Rules
    Triggers
  ride_request
  taxi
  user_roles
  users
  Trigger Functions
  Types

```

7. Challenges Faced

- Handling non-linear relationships in ETA prediction
- Preventing overfitting through proper train-test separation
- Ensuring feature consistency between training and prediction phases
- Integrating ML predictions into a secure backend workflow
- Managing complex relational data across multiple entities
- Generating accurate and meaningful AI-based ride summaries

8. Learnings & Skills Acquired

- Hands-on experience with **Spring Boot** backend development
- Practical knowledge of **PostgreSQL, JPA, and Hibernate**
- End-to-end implementation of **Random Forest** machine learning models
- Feature engineering for **geospatial and time-based data**
- Model evaluation using **R², RMSE, and MAE**
- ML model persistence and performance tracking
- Experience integrating **AI and ML systems into production backends**
- Improved REST API design, testing, and debugging skills

9. Conclusion

The NeuroFleetX project provided valuable hands-on experience in building a secure, scalable, and intelligent backend system integrated with a machine learning–based ETA prediction engine. Through this project, I strengthened my backend development skills, gained in-depth exposure to machine learning model development and evaluation, and learned how to deploy AI-driven features in real-world systems.

The Random Forest–based ETA model delivered accurate, robust, and explainable predictions, making the system suitable for real-world transportation and fleet management use cases. This internship significantly enhanced my technical confidence and prepared me for building enterprise-grade, AI-enabled applications.

10. Acknowledgements

I would like to express my sincere gratitude to **Infosys Springboard** for providing the opportunity to work on the NeuroFleetX project. I am thankful to my mentor **Vinay Prashant** sir for continuous guidance and support throughout the internship. I also appreciate my team members for their collaboration and teamwork, which contributed greatly to the successful completion of the project.